Syed Hashim Shah

Quantitative ML & Generative AI Researcher for Finance & Risk Management

syed.hashim.shahh@gmail.com linkedin.com/in/syed-hashim-shah Tokyo, Japan

Work Experience

Senior Consultant — Aurora Solutions **Tokyo, Japan**
Quantitative ML & Generative AI Researcher for Finance & Risk Management *Oct 2023 – Present*

- Research and deliver **quantitative ML and generative-AI solutions for finance and risk** — modelling and optimization in production Python.
- **Lead author** of **JPX Working Paper Vol. 48** — *Hypothetical Market Data Scenario Generation Using Generative AI* — for Japan Exchange Group / JSCC *(Aug 2025)*.
- Build **quantitative and statistical models** for risk and forecasting: **flow models, probabilistic time-series modelling, anomaly detection, and Bayesian methods**, with rigorous back-testing.
- Perform **time-series analysis and generation** — generative and probabilistic models for synthetic market-scenario data.
- Apply **optimization and Monte Carlo stochastic modelling** (convex optimization, portfolio and parameter optimization) to improve model accuracy and robustness.
- Build reproducible **data-analytics and feature-engineering pipelines** over large, noisy datasets.

Engineer and Researcher — GIKI, Faculty of Materials & Chemical Engineering **Topi, Pakistan**
Research & Teaching *Sep 2016 – Mar 2018*

- Established a new **Electrochemistry and Nanotechnology laboratory** and designed its undergraduate experiments from the ground up.
- Lecturer and instructor for *Introduction to Materials Engineering Lab* and *Advanced Materials Lab* to 400+ students; teaching assistant for Nanotechnology, Advanced Materials, and Thermodynamics.
- Conducted research on nanomaterials and electrochemical characterization of electrode materials.

Education

Kyoto University — Dept. of Biophysics, Graduate School of Science **Kyoto, Japan**
PhD in Computational Biophysics (research completed) *Oct 2020 – Present*

- All-atom molecular-dynamics simulations of proteins and DNA on supercomputers including **Fugaku**; free-energy calculations and enhanced-sampling methods.
- *Focus:* Data Science, Mathematical Optimization, Computational & Statistical Physics, Computational Biology, Python.

Nagoya University — Dept. of Physics, Graduate School of Science **Nagoya, Japan**
M.Sc. in Materials Physics (Theoretical Biophysics Lab) — GPA 4.00/4.00 *Oct 2018 – Sep 2020*

- Quantum-chemical enhanced sampling and free-energy calculations; Quantum Physics, Statistical Mechanics, Computational Physics.
- *Focus:* Markov Chain Monte Carlo, Optimization, Statistical Physics, Biophysics.

Ghulam Ishaq Khan Institute (GIKI) **Swabi, Pakistan**
B.Sc. in Materials Engineering (Nanotechnology) — GPA 3.88/4.00, Gold Medal *Aug 2012 – Jun 2016*

- *Focus:* Core Engineering, Mathematics, Physics, Nanomaterials, Battery Materials, Alloy Development, Testing.

Technical Skills

- **Programming & Tools:** Python, C++, Linux, Bash, Jupyter; data analytics, statistical computing, and scientific visualization.
- **Quantitative Methods:** Machine Learning, Statistical Modelling, Bayesian Inference, Modern Statistics, Optimization Algorithms, Graph Theory, Markov Chain Monte Carlo, Stochastic Processes, Time-Series Analysis, High-Dimensional Optimization.
- **Scientific Computing:** Molecular Dynamics (GROMACS, PLUMED, AMBER), Enhanced Sampling, Free-Energy Calculations, HPC / supercomputing (Fugaku).
- **Domain (Materials):** Nanomaterial synthesis, alloy development, electrochemical materials and testing.
- **Languages:** English (fluent), Japanese (**JLPT N3**), Urdu (native).

Awards & Achievements

- **MEXT Scholarship** — Fully funded Japanese Government Monbukagakusho Scholarship for Masters and PhD *(Apr 2018 – Sep 2023)*.
- **Lindau Nobel Laureate Meeting** — Fellowship from the Lindau Foundation and Else Kröner-Fresenius-Stiftung *(Jun 2023)*.
- **Merit Scholarship** — Fully funded award covering the entire undergraduate degree at GIKI *(2012)*.
- **Gold Medal** — Best academic performance in B.Sc. Materials Engineering, GIKI *(Jun 2016)*.
- **First Position** — Senior Design Project Competition, GIKI Industrial Open House *(Apr 2016)*.
- **Dean's Honour Roll** — All semesters, with the highest distinction.

Research Theses

Master's Thesis — Free-Energy Calculations of the Proline Dihedral by Quantum MD / Replica-Exchange Umbrella Sampling. Coupled quantum and classical molecular dynamics (AMBER, GAMESS) to compute the free-energy cost of the 180° proline transition with greater accuracy than classical MD alone, using Replica-Exchange Umbrella Sampling — quantifying a conformational flexibility important to protein structure.

Undergraduate Thesis — Design and Development of Supercapacitor Electrode Materials. Replaced expensive, heavy metallic foams with commercial-grade polyurethane foam combined with carbon nanotubes, polyaniline, and a cobalt-hydroxide nanocomposite, via a scalable wet-chemical route — achieving **680 F/g** of electrode weight in a cheaper, lighter supercapacitor.

Schools, Conferences & Publications

- Shah, S. H. (lead author) (2025). *Hypothetical Market Data Scenario Generation Using Generative AI.* **JPX Working Paper Vol. 48**, Japan Exchange Group (JPX) / JSCC.
- Shah, S. H., Khan, M. I., Sarfraz, R., Nazir, R., & Ali, A. (2018). *Polyaniline Enhanced Supercapacitance of Cobalt Hydroxide Nanowires/Carbon Nanotube Containing Polymer Sponge Layered Composite.* **Key Engineering Materials, 778, 175–180.**
- **Presentation & Poster:** *Elucidation of nucleosome sliding mechanism in all-atom detail via MD simulations* — 60th (Sep 2022) and 59th (Nov 2021) Annual Meetings, Biophysical Society of Japan.
- **Poster:** *Supercapacitors on CNT-coated polymeric foams* — 15th International Symposium on Advanced Materials, National Centre for Physics, Islamabad (2017).
- Asian Winter School, Institute of Molecular Science, Okazaki *(Jan 2020)*.
- Winter School on Quantum Chemistry, Institute of Molecular Science, Okazaki *(Dec 2019)*.
- 10th Toyota Riken International Workshop on *Science of Life Phenomena Woven by Water and Biomolecules*, Toyota Physical and Chemical Research Institute *(Sep 2019)*.

Continuous Professional Learning

- **Introduction to Corporate Finance** — Wharton School, University of Pennsylvania.
- **Python and Statistics for Financial Analysis** — Hong Kong University of Science and Technology.
- **Data Visualization with Python** — IBM.
- **Introduction to Stochastic Processes** — HSE University.